LENA VARGA-HOLM

Senior Data Engineer | Data Platforms

PROFESSIONAL SUMMARY

Senior Data Engineer with 9 years of experience designing, building and operating production data platforms and cloud data warehouses across logistics, payments and retail. Experienced with Python, SQL, Airflow, dbt, Kafka and Snowflake, including large-scale batch and streaming workloads. Focused on reusable data frameworks, dependable pipelines, data quality, clear data contracts and warehouse performance.

EXPERIENCE

Senior Data Engineer | Lukovac Freight Hub d.o.o. (logistics marketplace) | March 2022–Present

- Own ingestion and orchestration for the central data platform, maintaining about 140 Airflow DAGs loading from Postgres, MySQL, Salesforce and third-party APIs into Snowflake.
- Migrated about 300 stored procedures into dbt, building more than 420 models with tests; reduced the nightly warehouse build from 5 hours 10 minutes to 1 hour 35 minutes.
- Introduced JSON Schema data contracts and CI checks between product and data teams, reducing broken-pipeline incidents from about 9 per month to about 2 per month.
- Designed and built a Kafka event pipeline using Kafka Connect and Snowpipe Streaming, processing about 30 million shipment events per day with end-to-end latency under 2 minutes at p95; added dead-letter and replay tooling.
- Replaced hourly full-table extracts from Postgres with Debezium CDC into Kafka, reducing replica CPU during loads from about 70% to about 25%.
- Reduced monthly Snowflake spend from about 38,000 to about 24,000 through warehouse sizing, auto-suspend and removal of unused tables.
- Participate in the data platform on-call rotation, one week in six; mentor two mid-level engineers and run weekly data platform office hours.

Data Engineer | Kovalent Pay (payments startup) | September 2019–February 2022

- Built PySpark batch jobs on AWS EMR to reconcile bank settlement files against internal ledgers, processing about 1.2 TB of raw files from S3 daily.
- Reduced finance team's manual reconciliation work from about 3 days to about 4 hours per month-end.
- Designed a Redshift dimensional warehouse schema with more than 30 fact and dimension tables for finance and risk.
- Implemented Great Expectations data quality checks on critical finance tables and wrote an internal Python library for S3 and Glue Data Catalog access used by four teams.

Junior Data Engineer / BI Developer | Adria Retail Group | June 2017–August 2019

- Maintained SSIS packages and T-SQL ETL into an on-premises SQL Server warehouse.
- Migrated 60 nightly SSIS jobs to Python scripts scheduled with cron.
- Built Power BI reports for store operations across more than 40 stores.

SELECTED ENGINEERING CONTRIBUTIONS

- Contributed two merged pull requests to an open-source Airflow provider, fixing a pagination bug and adding tests.

- Presented “Data contracts without the hype” at a regional data meetup in 2024.
- Teach SQL for beginners at a local women-in-tech meetup, about four evenings per year since 2021.

EDUCATION

MSc Computer Science, Data Science track | Northgate Technical University | 2017
Thesis: Incremental materialized views for analytical queries

CERTIFICATION

AWS Certified Data Engineer – Associate | May 2024

TECHNICAL SKILLS

Languages: Python, SQL, Bash
Data platforms and engineering: Airflow, dbt, Spark (PySpark), Kafka, Kafka Connect, Debezium, Snowflake, Redshift, PostgreSQL, Great Expectations, JSON Schema, data contracts, dimensional modelling
Cloud and infrastructure: AWS (S3, EMR, Glue, Lambda, IAM, MSK), Docker, Terraform (basic), GitHub Actions
BI: Power BI, Looker (basic)

LANGUAGES

Croatian (native), English (C1), German (B1)